

SEONGSAN, Korea, Republic of

WMO: 471880

Lat: 33.3867N

Lon: 126.8803E

Elev: 64

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	31.9	17.2	13.1	37.7	19.4	14.6	37.8	18.3	46.6	16.1	43.1	7.1	320	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	88.7	80.3	87.1	79.9	85.4	79.1	81.9	86.2	80.9	85.2	79.9	84.0	9.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	160.6	84.4	79.7	154.6	83.7	78.6	148.9	83.1	45.9	86.2	44.7	85.2	43.6	84.4	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.6	16.0	14.1	DB	27.2	91.0	3.1	1.9	25.0	92.4	23.2	93.5	21.4	94.6	19.1	96.0
			WB	24.4	82.9	2.5	1.6	22.6	84.0	21.1	84.9	19.7	85.8	18.0	86.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	41.2	43.0	49.1	57.4	64.6	70.1	78.3	80.5	74.1	65.4	55.8	45.2	(6)
(7)		DBStd	14.13	5.53	6.13	5.48	4.47	3.40	2.68	4.09	3.54	3.15	4.93	6.90	6.17	(7)
(8)		HDD50	774	278	208	85	3	0	0	0	0	0	22	178		(8)
(9)		HDD65	3080	738	616	494	230	47	1	0	0	55	284	615		(9)
(10)		CDD50	4600	5	12	57	225	452	602	879	945	722	477	195	29	(10)
(11)		CDD65	1430	0	0	0	1	35	153	414	480	272	67	8	0	(11)
(12)		CDH74	9991	0	0	0	4	79	233	3493	4836	1229	114	3	0	(12)
(13)		CDH80	2471	0	0	0	0	1	4	852	1512	101	1	0	0	(13)
(14)	Wind	WSAvg	7.0	7.1	7.6	7.6	7.5	6.5	5.8	7.0	7.1	7.4	6.6	6.7		(14)
(15)	Precipitation	PrecAvg	71.9	2.9	3.6	4.6	5.5	6.6	9.4	10.9	11.3	8.0	3.5	3.3	2.2	(15)
(16)		PrecMax	125.7	6.6	8.1	10.6	8.3	13.0	33.1	27.5	30.8	29.2	16.5	7.6	7.0	(16)
(17)		PrecMin	42.1	0.7	1.4	1.3	2.0	0.5	1.6	3.2	2.6	1.5	0.0	0.5	0.2	(17)
(18)		PrecStd	17.9	1.7	2.0	2.4	1.7	3.6	6.5	6.9	7.0	6.4	3.8	1.9	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.1	61.9	65.9	71.4	79.2	80.0	90.3	91.0	84.5	78.9	73.8	66.1	(19)
(20)		MCWB	55.5	59.3	57.5	60.5	64.1	69.4	80.6	80.2	74.8	74.3	66.0	63.0		(20)
(21)		2%	DB	56.5	60.0	63.3	68.8	76.1	77.8	88.1	89.4	82.5	77.0	71.0	61.4	(21)
(22)		MCWB	51.3	56.8	57.0	60.1	62.5	70.5	80.3	80.4	74.4	73.0	65.7	56.4		(22)
(23)		5%	DB	53.2	56.0	61.5	67.0	73.9	76.4	86.3	88.1	81.0	75.1	68.7	58.5	(23)
(24)		MCWB	48.3	50.7	56.4	59.0	62.8	70.7	79.8	80.2	74.1	69.1	63.9	52.9		(24)
(25)		10%	DB	50.1	53.3	59.5	65.2	71.9	75.2	84.6	86.7	79.5	73.1	66.4	55.0	(25)
(26)		MCWB	44.7	47.4	54.7	58.6	63.0	70.6	79.0	79.7	73.1	66.8	61.0	49.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	60.7	61.3	65.0	70.0	76.1	83.2	83.7	78.7	75.9	69.3	63.6	(27)
(28)		MCD B	59.3	61.5	62.8	68.4	72.1	77.3	87.7	87.3	82.2	78.0	72.0	65.9		(28)
(29)		2%	WB	51.9	56.9	59.6	63.6	68.4	74.2	81.6	82.5	77.2	73.9	67.0	58.1	(29)
(30)		MCD B	55.5	58.9	61.7	66.0	71.5	75.7	86.1	86.6	80.1	76.4	69.9	60.6		(30)
(31)		5%	WB	48.9	52.4	58.1	62.2	67.1	73.0	80.5	81.5	76.0	71.3	64.5	53.6	(31)
(32)		MCD B	53.0	55.0	60.9	64.6	70.6	74.8	85.0	85.7	79.0	74.4	68.1	58.2		(32)
(33)		10%	WB	45.9	48.5	55.2	60.8	65.9	71.9	79.6	80.6	74.8	68.1	62.0	50.4	(33)
(34)		MCD B	49.9	52.3	59.2	63.6	69.5	74.1	83.4	84.8	78.1	71.6	65.7	54.9		(34)
(35)	Mean Daily Temperature Range		MDBR	9.5	10.6	12.7	12.9	12.2	8.2	7.8	8.5	8.8	10.2	11.1	9.9	(35)
(36)		5% DB	MCDBR	14.5	14.0	15.6	15.2	16.6	10.8	10.2	10.6	10.3	10.4	12.0	12.9	(36)
(37)		MCWBR	12.1	12.3	12.8	10.9	10.5	6.5	5.3	5.1	5.8	7.0	9.4	10.7	(37)	
(38)		5% WB	MCDBR	13.7	13.0	14.1	9.8	11.4	7.3	9.1	9.2	8.4	8.2	10.1	11.5	(38)
(39)		MCWBR	12.3	14.0	13.4	9.2	8.8	5.5	5.4	5.1	6.0	7.3	9.4	10.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.459	0.499	0.552	0.563	0.571	0.600	0.580	0.537	0.498	0.469	0.457	0.451	(40)	
(41)		taud	1.968	1.904	1.792	1.820	1.855	1.876	1.976	2.106	2.153	2.100	2.054	1.996	(41)	
(42)		Ebn at Noon	226	232	231	237	236	229	233	241	245	240	227	220	(42)	
(43)		Edh at Noon	46	55	66	67	65	64	57	50	46	44	42	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	665	926	1307	1553	1662	1448	1597	1629	1335	1178	826	619	(44)	
(45)		RadStd	76	113	103	128	191	179	228	165	129	117	94	53	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A

Nomenclature: See separate page